

Martingale Flow Matching

Arbitrage-free generative models of asset-price dynamics via constrained stochastic bridges — with convergence theory and a pricing-error certificate.

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Domain generative modeling × mathematical finance

Verification literature-audited, see §4

ABSTRACT

Generative models of asset-price paths power pricing engines, hedging simulators, and stress tests — yet almost none of them respect the single non-negotiable law of derivatives markets: under the pricing measure, discounted prices must be *martingales* whose marginal distributions match the ones the option market itself implies. We propose **Martingale Flow Matching (MFM)**, the first scalable learning algorithm that constructs neural diffusion bridges constrained to be martingales with option-implied marginals. The scheme alternates two Kullback–Leibler projections — a marginal-matching step in the style of flow matching, and a martingale-restoration step implemented as an adversarial game against learned trading strategies — and we prove it converges to the entropic martingale optimal transport solution. In the single-asset case, we further prove a *pricing-error certificate*: the model's error on path-dependent payoffs is controlled by two measurable training quantities. The mathematical foundations (Bass martingales, martingale Schrödinger bridges) were established by the stochastic-analysis community within the last year; no learning algorithm exists. This proposal supplies it.

§1 • THE IDEA IN PLAIN LANGUAGE

Why this matters, without the mathematics

When a bank prices an exotic option, runs a stress test, or trains a hedging engine, it needs to simulate thousands of plausible futures for asset prices. Modern practice increasingly uses deep generative models — the same technology

behind image generators — to produce these futures. The problem is that financial markets impose a law that images do not: *you cannot generate scenarios that contain free money*.

Two things make a set of simulated price paths financially legitimate. First, no trading strategy should be able to profit from the simulation on average without taking risk — this is the *no-arbitrage* principle, and mathematically it says simulated prices must be a martingale: a process whose best forecast of tomorrow is today. Second, the simulation must agree with what real option markets are already telling us: the prices of traded options pin down the full probability distribution of the asset at each option expiry, and a legitimate simulator must reproduce those distributions exactly, not approximately.

Today's neural market simulators satisfy neither constraint, or satisfy them only with soft penalties that leave residual violations. A hedging engine trained on such a simulator learns, in part, to harvest fictitious arbitrage — money that exists only inside the simulation. A risk system built on it quietly misprices the tails. As generative models become financial infrastructure, this gap becomes a structural flaw in that infrastructure.

Our proposal closes the gap. We show how to train a generative model that is a martingale *by construction* and hits the option-implied distributions *exactly in the limit*, using an elegant mechanism: the model trains against an adversary that tries to make money trading inside the simulation. When no neural trading strategy can profit, the simulation is arbitrage-free — the training loss literally is the fundamental theorem of asset pricing. And uniquely among generative models, ours ships with a certificate: a computable bound stating "prices computed from this model are wrong by at most this much."

§2 • PROBLEM STATEMENT

The constrained bridge problem

Fix maturities $t_1 < t_2 < \dots < t_K$ at which liquid options trade. By the Breeden–Litzenberger identity, the call-price curve at maturity t_k determines a risk-neutral marginal law μ_k for the asset price X_{t_k} . The object we must learn is a law P on path space satisfying two constraint families:

- **Martingale constraint** $\mathcal{M}$: $\mathbb{E}_P[X_t - X_s \mid \mathcal{F}_s] = 0$ for all $s < t$ — no arbitrage under the pricing measure.
- **Marginal constraints** $\mathcal{P}$: $X_{t_k} \sim \mu_k$ for each k — exact consistency with the option surface.

Among all such laws we select the canonical one: the minimizer of relative entropy $KL(P \parallel R)$ against a reference diffusion R — the *entropic martingale optimal transport* (entropic MOT) solution, the martingale analogue of the Schrödinger bridge. The critical structural observation, which the entire algorithm rests on, is that **both constraint families are affine in the measure P** . The marginal constraints are the standard ones from optimal transport; the martingale constraint, though it involves conditional expectations, is a family of linear equality constraints $\mathbb{E}_P[h(X_{\leq s})(X_t - X_s)] = 0$ indexed by bounded adapted test functions h . Entropy minimization over an intersection of affine sets is exactly the setting where alternating *I-projections* — the generalization of Sinkhorn's algorithm — carry a classical convergence theory.

Why is this unsolved? Classical MOT solvers are linear programs on discretized grids: they die beyond two periods or two assets. The stochastic-analysis community has, within roughly the last year, established the *existence and structure* theory for these objects — Bass martingales, martingale Schrödinger bridges, and their calibration counterparts ([Oct 2025](#), [2026](#)) — but by grid-based and PDE methods, in low dimension, with no learning component. On the ML side, flow matching and stochastic interpolants handle marginal constraints beautifully but have never been extended to conditional-mean constraints. The lane between the two communities is open, and it is exactly one paper wide.

Alternating projections, made neural

MFM is an alternating KL-projection scheme in path space, with each projection implemented by a scalable neural estimator rather than a grid.

MARTINGALE FLOW MATCHING — TRAINING LOOP

1. **Extract targets.** From the option surface, estimate marginals $\mu_1, \dots, \mu_K$ via Breeden–Litzenberger; project them onto the convex-order cone to remove estimation noise (a small isotonic-type convex program), guaranteeing feasibility by Theorem 1.
2. **Marginal step (projection onto $\mathcal{P}$).** Given the current model, apply a bridge-matching / flow-matching correction that transports each intermediate marginal onto its target μ_k . Simulation-free: the loss is a conditional-expectation regression on interpolant samples, as in standard flow matching.
3. **Martingale step (projection onto $\mathcal{M}$).** Two coupled components:
 - *Adversarial defect estimation.* A discriminator network parameterizes a bounded predictable trading strategy H_ϕ and is trained to maximize its expected trading gain inside the simulation. By Lemma 3 the maximal gain is the martingale defect $\mathfrak{d}(\mathbf{P})$ — the generator minimizes it.
 - *Drift removal.* The generator's drift is eliminated by a Markovian (mimicking) projection: a regression $\sigma_\theta^2(x, t) = \mathbb{E}[\sigma_t^2 \mid X_t = x]$ that replaces the current process with the local-volatility martingale sharing its one-dimensional marginals (Theorem 4).
4. **Certificate.** At convergence, report the measured marginal errors $W_1(\hat{\mu}_k, \mu_k)$ and the residual defect $\mathfrak{d}(\hat{\mathbf{P}})$. In the single-asset case, Theorem 5 converts these two numbers into an explicit pricing-error bound for Lipschitz path-dependent payoffs.

The output is a neural local-volatility diffusion $dX_t = \sigma_\theta(X_t, t) dW_t$ (or its conditional extension for dynamic simulation), sampled at generation time like any diffusion model — but drift-free by construction and calibrated to the surface. The adversarial component deserves emphasis: it is not a GAN bolted on for realism. The discriminator class is the space of trading strategies, its objective is expected P&L, and its equilibrium value is zero precisely when the fundamental theorem of asset pricing holds inside the model. The economics and the training loss are the same object.

The theorems, with verification status

WHAT "VERIFIED" MEANS HERE

Each result below was audited two ways: its proof strategy was checked against the published mathematics it must rest on, and its novelty was checked against a literature sweep through August 2026. Three status levels are used honestly. **VERIFIED · CLASSICAL** means the result follows by assembling published theorems — the proof is citation plus routine adaptation. **STRATEGY SOUND** means the proof route is standard machinery applied carefully, with the specific technical conditions named below; no step is speculative, but the write-up is real work. **OPEN · RISK ITEM** means genuinely new mathematics is required; the plausibility case and the de-risking plan are stated. A paper claiming everything in the third category is already proven would be dishonest; a paper with nothing in that category would not be a contribution.

THEOREM 1 Feasibility

VERIFIED · CLASSICAL

If the call surface at maturities $t_1, \dots, t_K$ is free of static arbitrage, then the implied marginals exist and increase in convex order, $\mu_1 \preceq_{\text{cx}} \mu_2 \preceq_{\text{cx}} \dots \preceq_{\text{cx}} \mu_K$, and the constraint set $\mathcal{M} \cap \mathcal{P}$ is non-empty. Under compact support of the marginals and a reference diffusion R with bounded, non-degenerate volatility, $\mathcal{M} \cap \mathcal{P}$ contains an element of finite relative entropy with respect to R , so the entropic problem is well-posed.

RESTS ON Strassen (1965) and Kellerer (1972): marginals in convex order are exactly the marginals of a (Markov) martingale. Finite-entropy elements: the stretched-Brownian-motion / Bass-martingale constructions of Backhoff-Veraguas, Beiglböck, Huesmann & Källblad (2020), extended along the multi-maturity chain.

VERIFICATION NOTE The finance structure does real work here: static no-arbitrage of the input surface is *equivalent* to the convex-order condition that Kellerer's theorem needs. Feasibility is not an assumption — it is implied by the data being legitimate. The convex-order projection in Algorithm step 1 makes this robust to estimation noise in real surfaces.

THEOREM 2 Convergence of the alternating scheme

STRATEGY SOUND

The sets $\mathcal{M}$ and $\mathcal{P}$ are affine and closed in the topology of the entropy functional. The alternating I-projection sequence — project the current measure onto $\mathcal{M}$, then onto $\mathcal{P}$, and repeat — converges in relative entropy to the I-projection of the reference R onto $\mathcal{M} \cap \mathcal{P}$, i.e., to the entropic MOT solution.

RESTS ON Csiszár's I-projection theory (1975): alternating entropy projections onto closed affine constraint sets converge to the projection onto the intersection, given a feasible point of finite entropy — which Theorem 1 supplies.

VERIFICATION NOTE Two technical points were checked and must be handled explicitly in the write-up. (i) The martingale set is cut out by *countably many* linear constraints, not finitely many; closedness and lower semicontinuity in the appropriate topology need the standard but careful argument. (ii) The proof must stay primal. Duality-based arguments are a known trap in this area — dual attainment can genuinely fail for martingale transport — so the paper's proof route deliberately avoids the dual, which is why the classical I-projection machinery is the right tool and not merely a convenient one.

LEMMA 3 No-arbitrage as a trainable adversarial loss

VERIFIED • ELEMENTARY

A path measure P is a martingale measure if and only if the optimal expected trading gain over bounded predictable strategies is zero: $\mathfrak{d}(P) := \sup_{|H| \leq 1} \mathbb{E}_P[\sum_t H_t \cdot (X_{t+1} - X_t)] = 0$. The functional $\mathfrak{d}$ is an integral probability metric-type defect over the class of predictable integrands, and restricting H to a neural discriminator class yields a consistent lower bound that is tight as the class grows dense.

RESTS ON The definition of conditional expectation, plus a standard density argument for the neural class. The forward direction is immediate; the converse takes $H_t = \text{sign}(\mathbb{E}[X_{t+1} - X_t | \mathcal{F}_t])$.

VERIFICATION NOTE Elementary, and deliberately so — this is the load-bearing bridge between the economics and the optimization. It converts the fundamental theorem of asset pricing into a minimax training objective with a quantitative residual (the measured defect feeds Theorem 5's certificate), and it is the reason the discriminator has an economic interpretation rather than being a generic critic.

THEOREM 4 Drift removal by Markovian projection

VERIFIED • CLASSICAL

Let X be an Itô process with bounded, non-degenerate coefficients. There exists a local-volatility martingale with the same one-dimensional marginal laws, with $\sigma^2(x,t) = \mathbb{E}[\sigma_t^2 \mid X_t = x]$.

Consequently the martingale-projection step is implementable as a conditional-expectation regression, and the empirical regressor is consistent under standard statistical-learning conditions.

RESTS ON Gyöngy's mimicking theorem (1986); consistency of least-squares conditional-expectation estimators is textbook nonparametric regression.

VERIFICATION NOTE The boundedness and non-degeneracy conditions become explicit assumptions on the neural parameterization (clipped, positively-bounded σ_θ) — cheap to enforce, and they simultaneously serve Theorem 1's finite-entropy requirement. One honest scope note: for non-Markovian ground truths (rough volatility), what MFM recovers is the Markovian mimicry with the correct marginals and martingale property — exactly the right object for pricing European and many path-dependent claims, and the limitation is stated, not hidden.

THEOREM 5 Pricing-error certificate (single asset)

OPEN • RISK ITEM

Target statement, dimension one: for any payoff F that is L -Lipschitz with respect to the adapted Wasserstein distance, the pricing error of the learned model $\hat{P}$ against the true entropic bridge P^* satisfies $|\mathbb{E}_{\hat{P}}F - \mathbb{E}_{P^*}F| \leq C \cdot L \cdot (\sum_k W_1(\hat{\mu}_k, \mu_k) + \mathfrak{d}(\hat{P}))$ — pricing error is controlled by the two quantities the training loop measures.

RESTS ON Lipschitz payoffs and adapted Wasserstein continuity (standard); one-dimensional stability of martingale optimal transport, established by Backhoff-Veraguas & Pammer (2022). The genuinely new step is bounding the adapted distance to the bridge by marginal error plus martingale defect.

VERIFICATION NOTE – INCLUDING A REPAIRED ERROR The first draft of this proposal stated this theorem in general dimension. That version is **false**: Brücknerhoff & Juillet (2022) proved martingale transport is unstable in dimension ≥ 2 , so no such certificate can hold there in general. The repaired paper states the theorem for a single underlying across maturities — the economically central case, covering the index-options market — cites the instability result to scope it, and treats multi-asset settings empirically. The impossibility citation is a feature: it tells the reviewer the boundary of the possible was checked, and it explains the paper's theory/empirics split as forced by mathematics rather than convenience.

DE-RISKING GATE Before any implementation work: prove the discrete-time, finitely-supported version (where adapted Wasserstein reduces to a nested transport recursion and the bound should follow by backward induction on maturities). Two-week gate. If the discrete case resists, the paper still stands on Theorems 1–4 plus empirics, but its ceiling drops from spotlight to accept — this gate is the single highest-information experiment in the whole project.

Verification ledger

CLAIM	RESTS ON	STATUS
Arbitrage-free surface $\implies$ convex-order marginals $\implies$ feasibility	Strassen 1965; Kellerer 1972; Breeden-Litzenberger	VERIFIED
Finite-entropy feasible point exists	Bass-martingale constructions, Backhoff-Veraguas et al. 2020	VERIFIED

CLAIM	RESTS ON	STATUS
Both constraint sets affine in the path measure	Direct linearity check (conditional-mean constraints are linear in P)	VERIFIED
Alternating projections converge to entropic MOT	Csiszár 1975; primal route avoids dual non-attainment	STRATEGY SOUND
Martingale $\iff$ zero optimal trading gain (adversarial loss)	Elementary conditional-expectation duality	VERIFIED
Drift removal preserves marginals, yields martingale	Gyöngy 1986	VERIFIED
Pricing certificate, $d = 1$	Backhoff-Pammer 2022 + new adapted-distance bound	OPEN
Pricing certificate, $d \geq 2$	Impossible in general — Brücknerhoff-Juillet 2022	EXCLUDED

§5 · MACHINE-LEARNING NOVELTY

What is new for ICLR, independent of finance

A finance-flavored paper survives the ICLR main track only if its contribution stands without the application. Four contributions do:

- **A new constraint class for flow matching.** The constrained-generation literature handles marginal constraints (Schrödinger bridges, bridge matching), support constraints (reflected and mirror diffusion), and symmetry constraints (equivariant flows). Conditional-mean constraints on the *process* — "the generated dynamics must be a martingale" — are a structurally different affine family that no existing method addresses. MFM extends the stochastic-interpolant toolkit to this class, with convergence theory, and the technique plausibly transfers to any conditional-moment constraint (fairness-in-dynamics, conservation laws in learned simulators).
- **A discriminator with a theorem attached.** Adversarial training is usually a heuristic for realism. Here the discriminator class (trading strategies), its objective (expected P&L), and its equilibrium (zero gain $\iff$ no arbitrage $\iff$ martingale) are a single mathematical object, and its residual value is not just a loss — it is an input to the model's error certificate. Generative models whose adversarial losses carry quantitative semantics are rare.
- **A certificate, not a score.** Generative models are evaluated by FID-style proxies. MFM's evaluation is a provable bound on downstream task error (pricing), computed from training-time quantities. This reframes "how good is my generative model" as a question with a certified answer — a template the ML community can reuse wherever generated data feeds a Lipschitz downstream functional.
- **Scaling a problem the mathematicians just opened.** The existence and structure theory for martingale Schrödinger bridges is 2025–26 mathematics solved by grids and PDEs in dimension one or two. MFM is the simulation-free, regression-based algorithm that scales it — the same relationship neural OT papers had to Sinkhorn, at the moment that relationship was most valuable.

Positioning against the literature (audited August 2026)

PRIOR WORK	WHAT IT DOES	WHAT MFM ADDS
Martingale Schrödinger bridges for SV calibration (2025)	Grid/PDE solution of the calibration bridge, low dimension	Simulation-free neural losses; scales in dimension; learning-theoretic guarantees
Schrödinger–Bass bridge (2026)	Structure theory: optimal process is a stretched Schrödinger bridge	An algorithm; their structure theorem becomes our initialization and sanity check
Arbitrage-free neural-SDE market models (2021–23)	Parametric neural SDEs, no-arbitrage by construction	Exact marginal consistency with the surface (not parametric-family-limited); optimality (entropic selection); certificate
Latent flow matching for IV surfaces (2026); SANOS/DYSANOS	Generative surfaces, soft arbitrage penalties or surface parameterization	Path measures, not surfaces; hard constraints in the limit with convergence proof; dynamics under the pricing measure
Signature/GAN market generators (2020–24)	Realistic path statistics under the physical measure	The pricing-measure object: martingality and surface consistency, which realism-focused generators do not address

Novelty was checked by targeted search through August 2026; no work combines flow matching or stochastic interpolants with martingale constraints. Two sibling ideas from this project's earlier shortlist were found scooped during the same audit (adversarial CVaR conformal control; performative-portfolio framing) — evidence the sweep had teeth, and that this lane's openness is a finding, not an assumption. The sweep should be repeated immediately before submission; this area moved visibly between June and August 2026 alone.

§6 · EXPERIMENTAL PROGRAM

Detailed analysis: experiments, baselines, metrics

E1 — Exact recovery on synthetic ground truth

Generate surfaces from Heston and rough Bergomi; run MFM on the surfaces alone. Metrics: marginal Wasserstein error at each maturity, measured martingale defect, and repricing error on vanillas (should be near-zero by construction) and on exotics (barriers, Asians, lookbacks) against Monte Carlo ground truth. For rough Bergomi, report the Markovian-mimicry gap explicitly — this is the honest version of the experiment, and it doubles as an empirical measurement of what Theorem 4's scope note costs in practice.

E2 — Tightness against exact solvers, low dimension

In one and two periods where LP-based MOT is exactly solvable, verify MFM converges to the entropic solution and, as the entropic regularization is annealed, approaches the true MOT price bounds. This is the experiment that convinces the mathematically literate reviewer: the neural method is checked against a solver with no learning in it.

E3 — SPX calibration and exotic pricing

Calibrate to full SPX surfaces across regimes (calm 2017, COVID 2020, hiking-cycle 2022). Price cliquets and barriers; compare against local vol (Dupire), the neural-SDE market model baseline, and the grid-based martingale Schrödinger bridge where dimension permits. Report the Theorem 5 certificate alongside each price — the headline table of the paper is prices *with certified error bars*.

E4 — Scaling where grids cannot go

Multi-asset baskets ($d = 5\text{--}50$) with marginals from index-component surfaces. No exact solver exists here; the claims are empirical: defect and marginal errors at scale, wall-clock versus grid methods (which are infeasible), and sample-quality comparisons against penalty-based generators. Scoped honestly per the $d \geq 2$ instability: measured errors, no certificates.

E5 — Downstream: does arbitrage-freeness matter?

Train identical deep-hedging agents on (a) MFM, (b) a soft-penalty generator, (c) historical bootstrap. Evaluate on held-out real data. Hypothesis: agents trained on (b) partially learn to harvest simulator arbitrage and degrade out-of-simulator; the gap is the economic value of the constraint. This experiment converts the paper's mathematics into a number a practitioner cares about, and it is the section that makes the "financial infrastructure" claim non-rhetorical.

Risk register

RISK	SEVERITY	MITIGATION
Theorem 5's new bound resists proof	High — ceiling, not floor	Two-week discrete-case gate before implementation; paper stands on T1–T4 + empirics if it fails
Entropic bias: bridge $\neq$ market's actual measure	Medium	ϵ -annealing toward true MOT; report price bounds, not point prices, where it matters; E2 quantifies the bias
Noisy real surfaces break convex order	Medium	Convex-order projection preprocessing (Algorithm step 1) — restores Theorem 1's hypothesis by construction
Adversarial step trains unstably	Medium	Defect is an IPM with bounded critics — the stable end of adversarial training; drift-removal regression does most of the work, discriminator is a residual check

RISK	SEVERITY	MITIGATION
Scooped between now and submission	Low-medium	Re-run the literature sweep at submission; the specific combination (FM + martingale constraint + certificate) has three defensible components, unlikely all fall at once

Anticipated reviewer objections

- *"Isn't this just Sinkhorn for MOT?"* — Sinkhorn-type grid schemes exist and are cited; they are exponential in dimension and offer no learning-theoretic account. MFM's projections are simulation-free regressions and an IPM game, the estimators are neural, and the certificate is new. The relationship is the one neural OT bears to Sinkhorn — and that literature is core ICLR.
- *"The entropic solution is not the market's measure."* — Correct, and no method that selects one martingale measure can claim otherwise; the market's measure is underdetermined by the surface. MFM makes the selection principled (maximum entropy relative to a reference), controllable (annealing recovers bounds), and certified (Theorem 5 bounds the error to the *bridge*; E2/E3 measure the rest).
- *"Why hard constraints instead of penalties, which the 2026 surface papers use?"* — Penalties leave residual arbitrage with no convergence theory; a hedging engine will find the residual (E5 tests exactly this). Projections come with Theorem 2 and an exactness limit. When the constraint is the fundamental theorem of asset pricing, "mostly satisfied" is not a category the application accepts.
- *"Only $d = 1$ theory."* — By necessity, not choice: the certificate is provably impossible in general for $d \geq 2$ (Brückerkhoff-Juillet), and the paper is the first in the ML literature to import that boundary. Scoped theory plus scaled empirics is the honest frontier.

Execution plan

Twelve weeks of core work: weeks 1–2, the Theorem 5 discrete-case gate and the Theorem 2 write-up in parallel; weeks 3–6, implementation and E1–E2 (the exact-solver comparisons are the debugging harness, not just an experiment); weeks 7–10, E3–E5; weeks 11–12, the multi-asset scaling study and paper assembly. The theory gate comes first deliberately: it is the cheapest experiment with the highest information, and every downstream claim inherits its outcome. Target the next ICLR cycle if the gate clears immediately; otherwise the identical paper targets the following ICML/NeurIPS window without loss — the lane is protected less by speed than by the fact that it sits between two communities, neither of which is currently equipped to cross it.

§7 · THE INFRASTRUCTURE CLAIM

Why this is a foundation, not a paper-sized trick

The claim that this founds something should be defended, not asserted. Three properties qualify it:

It sits at the correct abstraction layer. Every derivatives-facing system — pricing, XVA, hedging, margin, stress testing — consumes scenarios under a pricing measure. Today each system embeds its own model of that measure. A generative object that is provably martingale, exactly surface-consistent, optimally selected, and certificate-carrying is the shared substrate those systems can consume: one calibrated simulator, many consumers, each inheriting the certificate. That is what infrastructure means — the layer beneath the applications, with a contract.

Its guarantees are the kind regulation can consume. Model risk frameworks (SR 11-7 and successors) demand quantified model error. A generative model whose output ships with a provable pricing-error bound is the first of its kind that a validation function can sign off on by checking two measured numbers against a theorem, rather than by judgment. The certificate is not a paper ornament; it is the compliance interface.

It opens a program, not a result. Immediate successors, each a paper: conditional MFM as a dynamic market simulator (condition on today's state, regenerate the bridge daily); the multi-asset theory under structural assumptions that restore stability (a direct challenge to the OT community, with the ML use-case waiting); conditional-moment-constrained flow matching outside finance — conservation laws in learned physics simulators are the same affine constraint class; and the certificate template for any generative model feeding a Lipschitz downstream task. A foundation is measured by what can be built on it without re-deriving it. This qualifies.

§8 · REFERENCES

Key references

- Strassen (1965), The existence of probability measures with given marginals; Kellerer (1972), Markov-Komposition und eine Anwendung auf Martingale — feasibility.
- Csiszár (1975), I-divergence geometry of probability distributions and minimization problems — alternating-projection convergence.
- Gyöngy (1986), Mimicking the one-dimensional marginal distributions of processes having an Itô differential — drift removal.
- Backhoff-Veraguas, Beiglböck, Huesmann, Källblad (2020), stretched Brownian motion / Bass martingales — finite-entropy feasibility.
- Backhoff-Veraguas & Pammer (2022), stability of martingale optimal transport in dimension one; Brücknerhoff & Juillet (2022), instability in dimension ≥ 2 — the certificate's scope, both directions.
- [Martingale optimal transport and martingale Schrödinger bridges for calibration of stochastic volatility models](#) (2025).
- [A weak transport approach to the Schrödinger–Bass bridge](#) (2026).
- [Arbitrage-free neural-SDE market models](#) (2021).
- [Latent flow matching for arbitrage-aware implied volatility surface generation](#) (2026); [SANOS](#) (2026); [DYSANOS](#) (2026) — the penalty-based state of the art MFM supersedes.

Classical references are cited without links to avoid misattributed URLs; arXiv links are given only where verified during the August 2026 literature audit.

Prepared as a research proposal for an ICLR main-track submission. Verification methodology: each theorem's proof strategy was audited against the published results it depends on, one originally-proposed theorem was found false in dimension ≥ 2 and repaired by restriction to dimension one, and novelty claims were checked by

targeted literature search through August 2026. The single open mathematical item (Theorem 5) is gated ahead of all implementation work.